Misinformation Detector Using ML

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Abstract

- The abstract addresses the challenge of misinformation and fake news increasing rapidly on digital platforms. It proposes a model leveraging **Supervised *Machine learning (ML),NaturalLanguageProcessing (NLP),TF-IDF Vectorization, Model Training with PassiveAggressiveClassifier and text Classification*** to classify news articles as real or fake. showing that machine learning can accurately detect misinformation, with ensemble and deep learning models performing best.

Introduction

- The rise of digital platforms has made sharing information easier but also enabled the rapid spread of fake news. To tackle this, supervised machine learning models are used to detect misinformation automatically.These models are trained on labeled data to distinguish real and fake news.Natural Language Processing (NLP) techniques process the text for analysis.Through TF-IDF vectorization, textual data is converted into numerical form. A text classification algorithm, such as the PassiveAggressiveClassifier, then learns from these features. Thus, machine learning ensures faster and more reliable fake news detection in the digital era.

Problem Statement

- The rapid growth of digital media has led to an increase in the spread of fake news across online platforms. Manual verification of news content is time-consuming and ineffective for large-scale data. Misinformation can mislead the public and damage the credibility of online information sources. There is a need for an automated system that can accurately detect and classify fake news. This project aims to develop a machine learning model using NLP techniques to identify and prevent the spread of false information.

Proposed Methodology

- Use Supervised Learning with labeled dataset (Real/Fake).
- Preprocess text using NLP techniques (cleaning, tokenization).
- Apply TF-IDF Vectorization to convert text into numerical data.
- Train model using PassiveAggressiveClassifier. It identifies linguistic and contextual patterns that indicate false information.
- Evaluate performance using accuracy and confusion matrix.

Imlementation

- **Step 1:** Import and preprocess dataset.
- **Step 2:** Split data into training and testing sets.
- **Step 3:** Extract features using TF-IDF Vectorizer.
- **Step 4:** Train PassiveAggressiveClassifier model, which gives accuracy.
- **Step 5:** Predict and evaluate on test data.
- **Step 6:** UI interface : The provided Python code creates a simple graphical user interface (GUI) using the ipywidgets library to check a news article. The application uses a pre-trained machine learning model, loaded via the joblib library, to process the text and presumably classify it.

Result Analysis

- The Fake News Detection model was tested with multiple real-world news headlines. Each headline was passed through the trained classifier to predict whether it is Real or Fake.
- **For example:** Headline: "Court Forces Ohio To Allow Millions Of Illegally Purged Voters Back On The Rolls" → Predicted as Fake News.
- **Headline:** "Pressure Needed To Stop North Korea Crossing Red Nuclear Hurdle" → Predicted as Real News.
- The model successfully distinguished between authentic and misleading news based on the trained dataset.

Conclusion

- The conclusion emphasizes the social and technical need for fake news detection. The developed model shows promise but faces practical challenges—dataset imbalance, evolving writing styles, and the need for continuous updating. The section calls for future enhancements such as multimodal detection, explainable AI, and potentially blockchain verification for improved trust.

References

- Key cited works include foundational papers and benchmark datasets:
- Detection techniques: Ahmed et al. (2017), Shu et al. (2017), Conroy et al. (2015).
- Benchmark datasets: Wang (2017), FakeNewsNet, Kaggle datasets.
- Review works and emerging social/contextual detection methods.
- Shu, K., Sliva, A., Wang, S., Tang, J. and Liu, H. (2017). Fake news detection on social media: A data mining perspective. ACM SIGKDD Explorations Newsletter , 19 (1), 22-36. https://doi.org/10.1145/3137597.3137600.

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